

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Cryptography and Network Security
COURSE CODE: CSA51

COURSE OUTCOME COVERED

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 40%

QUESTIONS: 5

Total Marks:50

Annexure A

APPLICATION- BASED QUESTIONS

Code of Conduct:

*I Dhaanush Rajan- 192421162 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*



Signature of the student:

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation

APPLICATION- BASED QUESTIONS

1. Scenario: Event Communication Security: A student organization is preparing to share confidential event details with its core members using a simple classical encryption method. During testing, a message “MEET AT NOON” is transformed into an encoded version using a fixed shift pattern.

- Reconstruct how the encrypted message might appear if a shift of 4 positions is applied.
- Explain how an authorized member, aware of the encoding rule, would retrieve the original message.

ANSWER:

Scenario: Event Communication Security

The student organization uses a **Caesar Cipher**, a classical substitution encryption technique, with a **shift of 4**.

a) Encryption of "MEET AT NOON" using a shift of 4

Each letter is shifted **4 positions forward** in the alphabet.

Plain Letter	Shift +4	Cipher Letter
M	Q	Q
E	I	I
E	I	I
T	X	X
A	E	E
T	X	X
N	R	R
O	S	S
O	S	S
N	R	R

Plaintext:
MEET AT NOON

Ciphertext:
QIIX EX RSSR

Encryption Process

M → Q
E → I
E → I
T → X

A → E
T → X

N → R
O → S
O → S
N → R

Encrypted Message:
QIIX EX RSSR

b) How an Authorized Member Retrieves the Original Message (Decryption)

The authorized member knows that the message was encrypted using a **shift of 4**. To decrypt it, each letter is shifted **4 positions backward**.

Cipher Letter	Shift -4	Plain Letter
Q	M	M
I	E	E
I	E	E
X	T	T
E	A	A
X	T	T
R	N	N
S	O	O
S	O	O
R	N	N

Ciphertext:
QIIX EX RSSR

Decrypted Plaintext:
MEET AT NOON

Algorithm

Encryption

Choose a shift value (**4**).

Replace each letter with the letter **4 positions ahead** in the alphabet.

Keep spaces unchanged.

The resulting text is the ciphertext

Decryption

Read the ciphertext.

Shift each letter **4 positions backward**.

Keep spaces unchanged.

The original plaintext is recovered

Final Answer

Plaintext: MEET AT NOON

Shift Value: 4

Ciphertext: QIIX EX RSSR

Decrypted Text: MEET AT NOON

This demonstrates how the **Caesar Cipher** encrypts a message by shifting letters forward and decrypts it by shifting the same number of positions backward.

2. Scenario: Intercepted Suspicious Message: A cybersecurity intern intercepts the coded message: “GSRH RH Z HVXIVG” during network monitoring. The structure of the text suggests a simple substitution mechanism rather than complex encryption.

- a. Deduce the most likely original message.
- b. Justify your answer by analysing repeating patterns, letter structures, or known cipher characteristics.

ANSWER

Scenario: Intercepted Suspicious Message

Ciphertext:

GSRH RH Z HVXIVG

The message is most likely encrypted using the **Atbash Cipher**, a classical substitution cipher in which each letter is replaced by its opposite in the alphabet:

- A ↔ Z
- B ↔ Y
- C ↔ X
- D ↔ W
- E ↔ V
- ...
- M ↔ N

a) Original Message

Applying the Atbash cipher:

Cipher Letter Plain Letter

G	T
S	H
R	I
H	S
R	I
H	S
Z	A
H	S
V	E
X	C
I	R
V	E
G	T

Ciphertext:

GSRH RH Z HVXIVG

Plaintext:
THIS IS A SECRET

b) Justification

The intercepted message is likely encrypted using the **Atbash Cipher** because:

1. Simple Substitution Pattern

- Every ciphertext letter consistently maps to a single plaintext letter.
- Example:
 - $G \rightarrow T$
 - $H \rightarrow S$
 - $R \rightarrow I$

2. Repeating Letter Patterns

- The repeated word **RH** appears twice.
- After decryption:
 - $R \rightarrow I$
 - $H \rightarrow S$
- So **RH** = **IS**, which is a common English word.

3. Recognizable Word Structure

- The single letter **Z** decrypts to **A**, forming the phrase:
 - **THIS IS A ...**
- The last word **HVXIVG** decrypts to **SECRET**, a meaningful English word.

4. Known Characteristics of Atbash Cipher

- It is a **monoalphabetic substitution cipher**.
- It uses a **reversed alphabet** instead of a shifted alphabet (unlike the Caesar cipher).
- No encryption key is required; the mapping is fixed.

Plain A B C D E F G H I J K L M
Cipher Z Y X W V U T S R Q P O N

3. Scenario: Secure Messaging System: A company deploys a keyword-based encryption technique for internal communication. An employee sends the word “HELLO” using the keyword “KEY” to ensure confidentiality.

- Demonstrate how the encryption process transforms the message step-by-step.
- Explain how the repeating keyword influences each character of the ciphertext.

ANSWER

Scenario: Secure Messaging System

The company uses the **Vigenère Cipher**, a classical keyword-based substitution cipher. In this method, each letter of the plaintext is shifted according to the corresponding letter of a **repeating keyword**.

- Plaintext:** HELLO
- Keyword:** KEY

The keyword is repeated until it matches the length of the plaintext:

- Plaintext: **H E L L O**
- Keyword: **K E Y K E**

a) Step-by-Step Encryption

Assign numerical values to the letters (A = 0, B = 1, ..., Z = 25).

Plain Letter Value Keyword Letter Value Calculation (Mod 26) Cipher Letter

H	7	K	10	$(7 + 10) = 17$	R
E	4	E	4	$(4 + 4) = 8$	I
L	11	Y	24	$(11 + 24) = 35 \rightarrow 9$	J
L	11	K	10	$(11 + 10) = 21$	V
O	14	E	4	$(14 + 4) = 18$	S

Encrypted Message

Plaintext: HELLO

Keyword: KEYKE

Ciphertext: RIJVS

b) How the Repeating Keyword Influences the Ciphertext

The keyword "**KEY**" is repeated to match the length of the plaintext:

Plaintext : H E L L O

Keyword : K E Y K E

Ciphertext: R I J V S

- Each **keyword letter determines the shift** for the corresponding plaintext letter.
- Different keyword letters produce **different shift values**, making the encryption stronger than a Caesar cipher.
- After reaching the end of the keyword, it **repeats** ($K \rightarrow E \rightarrow Y \rightarrow K \rightarrow E \dots$) until the entire message is encrypted.
- Because the shift changes from one character to the next, the same plaintext letter can encrypt to **different ciphertext letters** depending on the keyword letter used.

For example:

- $L + Y \rightarrow J$
- $L + K \rightarrow V$

Although both plaintext letters are **L**, they become **different ciphertext letters** because they are paired with different keyword letters.

Final Answer

- **Plaintext:** HELLO
- **Keyword:** KEY
- **Repeated Keyword:** KEYKE
- **Ciphertext:** RIJVS

The Vigenère cipher enhances security by using a **repeating keyword** to apply a different shift to each character, making frequency analysis more difficult than with a simple Caesar cipher.

4. Scenario: Suspicious Digital Asset Investigation: During a digital forensic audit, investigators suspect that a company logo file may contain hidden confidential information embedded within it. The file appears normal upon casual inspection.

- a. Analyze how investigators could detect the presence of hidden data within the image.
- b. Recommend two techniques that could strengthen the security of such hidden communications against detection or extraction.

ANSWER

Scenario: Suspicious Digital Asset Investigation

Investigators suspect that a company logo contains hidden confidential information using **steganography**, where secret data is embedded inside an image without visibly changing its appearance.

a) How Investigators Can Detect Hidden Data

Investigators can use the following forensic techniques:

1. Visual Inspection

- Compare the suspected image with the original version.
- Zoom in to identify unusual pixels, distortions, or color variations.
- Although hidden data is usually invisible, slight anomalies may appear after enhancement.

2. Statistical (Steganalysis) Analysis

- Analyze pixel value distributions and image histograms.
- Hidden data often alters the **Least Significant Bits (LSBs)** of pixel values.
- Statistical tests can reveal abnormal patterns that indicate embedded information.

3. Metadata Examination

- Check image metadata (EXIF information).
- Unexpected software names, editing history, or unusual timestamps may indicate steganographic processing.

4. File Size Comparison

- Compare the suspected image with the original.
- A larger-than-expected file size may suggest that hidden data has been embedded.

5. Steganalysis Tools

Use forensic software such as:

- StegExpose
- StegDetect
- OpenStego
- Steghide
- Binwalk

These tools scan images for hidden payloads and attempt to extract embedded information.

b) Two Techniques to Strengthen Hidden Communications

1. Encrypt the Secret Data Before Embedding

- Encrypt the confidential message using algorithms such as **AES** before hiding it inside the image.
- Even if an attacker extracts the hidden data, it remains unreadable without the decryption key.

Benefit:

- Provides an additional layer of confidentiality.

2. Use Adaptive or Randomized Steganography

- Instead of embedding data sequentially, hide it in randomly selected pixels using a secret key.
- Adaptive methods choose image regions with more texture or noise, making changes harder to detect through statistical analysis.

Benefit:

- Reduces the likelihood of steganalysis detecting the hidden message.
- Makes extraction significantly more difficult without the embedding key.

Final Answer

a) Detection Methods

- Visual inspection of the image.
- Statistical (steganalysis) analysis of pixel values and histograms.
- Metadata examination.
- File size comparison.
- Use of specialized steganography detection tools.

b) Security Enhancement Techniques

1. **Encrypt the secret message before embedding** (e.g., using AES).
2. **Use adaptive/randomized steganography** with a secret embedding key to make detection and extraction much more difficult.

Conclusion: Combining **encryption** with **advanced steganography techniques** provides stronger protection for hidden communications by ensuring both confidentiality and resistance to forensic detection.

5. Scenario: Legacy Encryption in Corporate Communication: A legacy system in a company still uses a transposition-based encryption method with three levels of arrangement. A received message appears as: “WECRLTEERDSOEFEAOCAIVDEN”.

- a. Determine the most probable original message by reversing the encryption process.
- b. Critically evaluate why such a method is vulnerable in modern communication systems.
- c. Propose one practical enhancement to improve the confidentiality of messages in this scenario.

ANSWER

Scenario: Legacy Encryption in Corporate Communication

The given ciphertext:

Ciphertext:

WECRLTEERDSOEFEAOCAIVDEN

This is a well-known example of the **Rail Fence Cipher** with **3 rails (levels)**.

a) Determine the Original Message

Step 1: Arrange the ciphertext into 3 rails

For a 3-rail Rail Fence Cipher, the zigzag pattern is:

Rail 1: W . . . E . . . C . . . R . . . L . . . T . . . E
Rail 2: . E . R . D . S . O . E . E . F . E . A . O . C .
Rail 3: . . A . . . I . . . V . . . D . . . E . . . N

Step 2: Read the zigzag pattern

Following the zigzag path row by row reconstructs the original plaintext:

Original Message:

WE ARE DISCOVERED FLEE AT ONCE

(Removing spaces gives:)

WEAREDISCOVEREDFLEEATONCE

b) Why is this Method Vulnerable in Modern Communication Systems?

The Rail Fence Cipher is considered insecure for modern communication because:

1. No Character Substitution

- The letters themselves are **not changed**.
- Only their positions are rearranged.

2. Easy to Break

- An attacker can try a small number of rail values (2, 3, 4, etc.).
- The correct plaintext is quickly recovered using brute force.

3. Preserves Letter Frequency

- Since the letters remain unchanged, frequency analysis is still effective.
- Common words and patterns are easily identified.

4. No Secret Key Complexity

- Security depends only on the number of rails.
- This provides a very small key space, making exhaustive search practical.

5. Unsuitable for Modern Threats

- Modern computers can decrypt Rail Fence ciphers almost instantly.
- It provides virtually no protection against current cryptographic attacks.

c) Practical Enhancement

Encrypt the message using a modern encryption algorithm such as **AES (Advanced Encryption Standard)** before transmission.

Advantages:

- Uses strong cryptographic keys (128, 192, or 256 bits).
- Resistant to brute-force attacks.
- Provides high confidentiality for sensitive corporate communications.
- Widely adopted as an industry standard.

Alternatively, if the Rail Fence Cipher must be retained for legacy compatibility, **combine it with a strong encryption algorithm (e.g., AES)** so that even if the transposition is reversed, the recovered data remains encrypted.

a) Original Message

WE ARE DISCOVERED FLEE AT ONCE

b) Vulnerabilities

- Only rearranges letters; does not change them.
- Small key space makes brute-force attacks easy.

- Preserves letter frequencies.
- Weak against modern cryptanalysis.
- Inadequate for protecting confidential communications.

c) Recommended Enhancement

Use **AES encryption** (or encrypt the message with AES before applying the Rail Fence Cipher). This significantly improves confidentiality and protects the message against modern attacks.
